

ide and Talamas (2015)

1. Setup

$$\begin{cases} \bar{z}_L < \bar{z}_H \\ \bar{z}_{AI} = a \end{cases} \quad \bar{z}_L < a < \bar{z}_H$$

$$x \sim U[0,1] \quad x \leq \bar{z}_L \quad 1 - \bar{z}_L$$

$$h n_L (1 - \bar{z}_L) = 1 \quad \rightarrow \quad n_L = \frac{1}{h(1 - \bar{z}_L)}$$

2. Autonomous AI

$$r^A = a \quad y_L^A = n_L a$$

$$n_L w_L^A + r^A$$

$$n_L a - n_L w_L^A - r^A = 0$$

$$n_L (a - w_L^A) = a$$

$$w_L^A = a \left(1 - \frac{1}{n_L} \right)$$

$$\frac{1}{n_L} = a \left(1 - \frac{1}{n_L} \right)$$

$$\frac{1}{n_L} = h \left(1 - \frac{1}{n_L} \right) \rightarrow w_L^A = a (1 - h(1 - \bar{z}_L))$$

$\frac{a}{n_L} \rightarrow$ is the part of output that must compensate the AI for its opportunity cost: an autonomous AI could instead produce independently

3. Nonautonomous AI

$$r^N = 0$$

$$y_L^N = n_L a$$

$$n_L a - n_L w_L^N - r^N = 0$$

$$r^N = 0$$

$$n_L (a - w_L^N) = 0$$

$$w_L^N = a$$

$$w_L^N - w_L^A = ah(1 - \bar{z}_L) > 0$$

$$r^A = a > 0, \quad r^N = 0$$

4. Comparing the low type

$$w_L^N - w_L^A = a - a \left(1 - \frac{1}{n_L} \right)$$

$$w_L^N - w_L^A = \frac{a}{n_L} > 0$$

$$w_L^N > w_L^A \quad \frac{1}{n_L} = h(1 - \bar{z}_L)$$

5. When is nonautonomous AI adopted?

$$w_L^0$$

$$w_L^N = a$$

$$w_L^N > w_L^0$$

$$a > w_L^0$$

6. total output

$$f^N \leq f^A$$

$$y^N = \max_{f \in f^N} y(f)$$

$$y^A = \max_{f \in f^A} y(f)$$

$$y^A \geq y^N, \quad a > 0$$

7. Main result

$$w_L^N = a > a \left(1 - \frac{1}{n_L} \right) = w_L^A$$

$$y^A > y^N$$

Nonautonomous AI benefits the low type more, while autonomous AI generates more total output.